

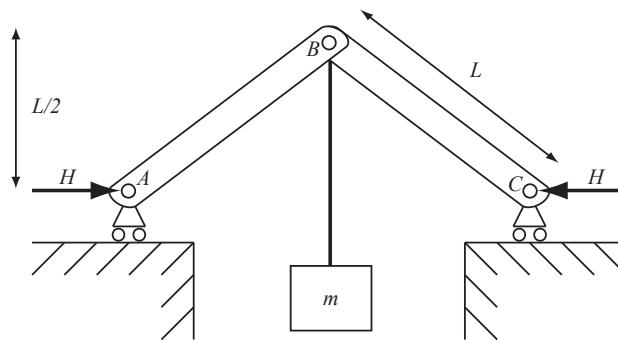
SECTION B

7 (**short**) The frictionless mechanism ABC shown in Fig. 5 is used to lift a body of mass m connected to the pin joint at B by a light, inextensible cable. The mechanism consists of two light, rigid bars AB and BC , each of length L .

For the case in which A and C each move with absolute speed $v/2$ in the direction of the applied forces H :

(a) Sketch the velocity diagram. [5]

(b) Determine the force H and the vertical support reactions V . [5]



(not to scale)

Fig. 5

8 (**short**) A tractor of mass $2m$ attempts to tow a small car of mass m on a frictionless horizontal road by means of a light, elastic cable. The cable is initially slack as the tractor starts to move with speed v . The cable then goes taut and the car begins to move.

(a) Calculate the elastic potential energy in the cable at the point of maximum extension of the cable. [8]

(b) Briefly explain what will happen to the elastic potential energy in the cable if the cable snaps at the point of maximum extension of the cable. [2]